

Supporting Information for:
Quantum-Coherent Spectral Engineering:
Non-Markovian Dynamics in Photosynthetic Energy
Transfer under Engineered Photon Baths

Steve Cabrel Teguia Kouam, Theodore Goumai Vedekoi, Jean-Pierre Tchapel Njafa,
Jean-Pierre Nguenang, and Serge Guy Nana Engo

March 25, 2026

Contents

1 PT-HOPS/SBD Formalism Details

1.1 Hierarchy Equations

The Process Tensor Hierarchy of Pure States (PT-HOPS) method extends the standard HOPS formalism by incorporating a process tensor that efficiently captures non-Markovian environmental memory. The hierarchy of pure states $|\psi^{(\vec{n})}(t)\rangle$ evolves according to:

$$\frac{\partial}{\partial t}|\psi^{(\vec{n})}(t)\rangle = \left(-i\mathbf{H}_S - \sum_j n_j \nu_j\right) |\psi^{(\vec{n})}(t)\rangle - i \sum_j \sqrt{(n_j + 1)d_j} L_j |\psi^{(\vec{n}+\vec{e}_j)}(t)\rangle - i \sum_j \sqrt{n_j/d_j} L_j^\dagger |\psi^{(\vec{n}-\vec{e}_j)}(t)\rangle, \quad (1)$$

where:

- $\vec{n} = (n_1, n_2, \dots, n_K)$ is the multi-index for hierarchy level
- ν_j are the bath correlation frequencies from Padé decomposition
- d_j are the decomposition coefficients
- L_j are the system coupling operators
- $\vec{e}_j$ is the unit vector in direction j

1.2 Bath Correlation Function Decomposition

The bath correlation function is decomposed using Padé spectral decomposition:

$$C(t) = \sum_{k=0}^K d_k e^{-\nu_k t}, \quad (2)$$

where K is the number of Matsubara terms. For the Drude–Lorentz spectral density:

$$J_{\text{DL}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \quad (3)$$

the correlation function at temperature T is:

$$C(t) = \frac{\lambda\gamma}{\beta} \sum_{k=0}^{\infty} \frac{e^{-\nu_k t}}{\nu_k - \gamma} + \frac{\lambda\gamma}{2} \left[\coth\left(\frac{\beta\gamma}{2}\right) - i \right] e^{-\gamma t}, \quad (4)$$

with Matsubara frequencies $\nu_k = 2\pi k/\beta$ and $\beta = 1/(k_B T)$.

1.3 Spectrally Bundled Dissipators

The SBD approach groups environmental modes by spectral frequency, reducing the hierarchy dimension. For M spectral bundles:

$$J_{\text{bath}}(\omega) \approx \sum_{m=1}^M J_m(\omega), \quad (5)$$

where each bundle $J_m(\omega)$ is treated as an independent bath with its own hierarchy. This enables scalable simulations of complex multisite systems.

Table 1: **PT-HOPS/SBD Simulation Parameters**

Parameter	Value
Hierarchy depth (L)	6
Matsubara terms (K)	3
Time step (Δt)	0.1 fs
Total simulation time	1 000 fs
Disorder realizations	100
Precision	Double (64-bit)

1.4 Numerical Parameters

2 FMO Hamiltonian Parameters

2.1 Site Energies

The site energies ε_n for the seven bacteriochlorophyll-*a* molecules in FMO (from Ref. [?]):

Table 2: **FMO Site Energies (cm^{-1})**

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12400	12550	12250	12350	12450	12600	12500

2.2 Electronic Couplings

The electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.9 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 1.2 & -6.3 & 1.7 \\ 6.7 & 0.7 & -2.2 & 1.2 & 0 & 9.6 & -6.0 \\ -13.7 & -11.9 & -9.6 & -6.3 & 9.6 & 0 & 89.7 \\ -9.9 & 4.3 & 6.0 & 1.7 & -6.0 & 89.7 & 0 \end{pmatrix}. \quad (6)$$

2.3 Spectral Density Parameters

3 12-Test Validation Suite

We implement a comprehensive 12-test validation suite organized in three categories to ensure observed quantum advantages are genuine physical effects rather than numerical artifacts.

3.1 Convergence Tests (4 tests)

Test 1: Hierarchy Depth Convergence

Purpose: Verify that hierarchy depth $L = 6$ is sufficient.

Method: Compare ETE for $L = 4, 5, 6, 7, 8$.

Acceptance criterion: $|\text{ETE}_{L=6} - \text{ETE}_{L=8}|/\text{ETE}_{L=8} < 2\%$.

Result: 1.2% deviation $\rightarrow$ PASS.

Test 2: Matsubara Terms Convergence

Purpose: Verify that $K = 3$ Matsubara terms capture thermal effects.

Table 3: Spectral Density Parameters

Parameter	Value
Drude–Lorentz reorganization (λ_{DL})	35 cm^{-1}
Drude–Lorentz cutoff (γ_{DL})	50 cm^{-1}
Vibronic mode 1 (ω_1)	150 cm^{-1}
Vibronic mode 2 (ω_2)	200 cm^{-1}
Vibronic mode 3 (ω_3)	575 cm^{-1}
Vibronic mode 4 (ω_4)	$1\,185 \text{ cm}^{-1}$
Huang–Rhys S_1	0.05
Huang–Rhys S_2	0.02
Huang–Rhys S_3	0.01
Huang–Rhys S_4	0.005
Temperature (T)	295 K

Method: Compare coherence lifetime for $K = 2, 3, 4, 5$.

Acceptance criterion: $|\tau_{c,K=3} - \tau_{c,K=5}|/\tau_{c,K=5} < 2\%$.

Result: 0.8% deviation $\rightarrow$ PASS.

Test 3: Time Step Convergence

Purpose: Verify that $\Delta t = 0.1 \text{ fs}$ is sufficient.

Method: Compare population dynamics for $\Delta t = 0.05, 0.1, 0.2 \text{ fs}$.

Acceptance criterion: Maximum population deviation $< 1\%$.

Result: 0.3% deviation $\rightarrow$ PASS.

Test 4: HEOM Benchmark

Purpose: Validate PT-HOPS against numerically exact HEOM.

Method: Compare ETE for 3-site system with known HEOM solution.

Acceptance criterion: $|\text{ETE}_{\text{PT-HOPS}} - \text{ETE}_{\text{HEOM}}|/\text{ETE}_{\text{HEOM}} < 2\%$.

Result: 1.8% deviation $\rightarrow$ PASS.

3.2 Physical Consistency Tests (4 tests)

Test 5: Trace Preservation

Purpose: Verify probability conservation.

Method: Monitor $\text{Tr}[\rho(t)]$ throughout simulation.

Acceptance criterion: $|\text{Tr}[\rho(t)] - 1| < 10^{-12}$ for all t .

Result: Maximum deviation $4.2 \times 10^{-13} \rightarrow$ PASS.

Test 6: Positivity Preservation

Purpose: Verify density matrix remains positive semi-definite.

Method: Check eigenvalues of $\rho(t)$ at each time step.

Acceptance criterion: All eigenvalues $\geq -10^{-14}$.

Result: Minimum eigenvalue $-2.1 \times 10^{-15} \rightarrow$ PASS.

Test 7: Detailed Balance

Purpose: Verify thermal equilibrium is correctly reproduced.

Method: Compare steady-state populations with Boltzmann distribution.

Acceptance criterion: $|p_n^{\text{steady}} - p_n^{\text{Boltzmann}}|/p_n^{\text{Boltzmann}} < 5\%$.

Result: Maximum deviation 2.3% $\rightarrow$ PASS.

Test 8: Hermiticity Preservation

Purpose: Verify density matrix remains Hermitian.

Method: Check $\rho = \rho^\dagger$ at each time step.

Acceptance criterion: $\|\rho - \rho^\dagger\|_F < 10^{-12}$.

Result: Frobenius norm $3.7 \times 10^{-14} \rightarrow$ PASS.

3.3 Environmental Robustness Tests (4 tests)

Test 9: Temperature Stability

Purpose: Verify coherence enhancement persists under temperature variations.

Method: Compute η at $T = 285, 295, 305$ K.

Acceptance criterion: $\eta > 0.15$ for all temperatures.

Result: $\eta = \{0.23, 0.25, 0.21\} \rightarrow \text{PASS}$.

Test 10: Static Disorder Robustness

Purpose: Verify coherence enhancement persists under energetic disorder.

Method: Ensemble average over 100 disorder realizations with $\sigma = 50 \text{ cm}^{-1}$.

Acceptance criterion: $\langle \eta \rangle > 0.15$ with $\text{CV} < 25 \%$.

Result: $\langle \eta \rangle = 0.20(4)$, $\text{CV} = 20 \%$ $\rightarrow \text{PASS}$.

Test 11: Bath Parameter Fluctuations

Purpose: Verify robustness to spectral density variations.

Method: Vary λ, γ by $\pm 20 \%$.

Acceptance criterion: $\eta > 0.15$ for all variations.

Result: $\eta \in [0.18, 0.26] \rightarrow \text{PASS}$.

Test 12: Markovian Limit Recovery

Purpose: Verify correct high-temperature limit.

Method: Compare with Redfield theory at $T = 500 \text{ K}$.

Acceptance criterion: $|\text{ETE}_{\text{PT-HOPS}} - \text{ETE}_{\text{Redfield}}|/\text{ETE}_{\text{Redfield}} < 5 \%$.

Result: 2.1 % deviation $\rightarrow \text{PASS}$.

Table 4: **12-Test Validation Summary**

Category	Test	Result	Status
Convergence	Hierarchy depth	1.2 %	PASS
	Matsubara terms	0.8 %	PASS
	Time step	0.3 %	PASS
	HEOM benchmark	1.8 %	PASS
Physical Consistency	Trace preservation	4.2×10^{-13}	PASS
	Positivity	-2.1×10^{-15}	PASS
	Detailed balance	2.3 %	PASS
	Hermiticity	3.7×10^{-14}	PASS
Environmental Robustness	Temperature stability	$\{0.23, 0.25, 0.21\}$	PASS
	Static disorder	0.20(4)	PASS
	Bath fluctuations	[0.18, 0.26]	PASS
	Markovian limit	2.1 %	PASS

4 Convergence Analysis

4.1 Hierarchy Depth

4.2 Matsubara Terms

5 Computational Cost Breakdown

All simulations performed on AMD Ryzen 5 5500U (6 cores, 12 threads) with 40 GB RAM.

Table 5: **Hierarchy Depth Convergence**

L	ETE	Deviation (%)	CPU Time (h)
4	1.221	2.4	12.3
5	1.238	1.0	28.7
6	1.250	—	52.1
7	1.256	0.5	89.4
8	1.262	1.0	142.6

Table 6: **Matsubara Terms Convergence**

K	τ_c (fs)	Deviation (%)	CPU Time (h)
1	398	5.2	31.2
2	412	1.9	42.8
3	420	—	52.1
4	423	0.7	61.5
5	425	1.2	70.9

6 Additional Quantum Metrics

6.1 Von Neumann Entropy

$$S_{\text{vN}}(t) = -\text{Tr}[\boldsymbol{\rho}(t) \ln \boldsymbol{\rho}(t)]. \quad (7)$$

Under filtered illumination: $S_{\text{vN}}(t = 500 \text{ fs}) = 0.51(6)$.

Under broadband illumination: $S_{\text{vN}}(t = 500 \text{ fs}) = 0.73(7)$.

Lower entropy indicates more ordered quantum state (beneficial for transport).

6.2 Purity

$$P(t) = \text{Tr}[\boldsymbol{\rho}(t)^2]. \quad (8)$$

Under filtered illumination: $P(t = 500 \text{ fs}) = 0.82(4)$.

Under broadband illumination: $P(t = 500 \text{ fs}) = 0.71(5)$.

Higher purity indicates less decoherence.

6.3 Mandel Q Parameter

$$Q_{\text{Mandel}} = \frac{\langle (\Delta n)^2 \rangle - \langle n \rangle}{\langle n \rangle}. \quad (9)$$

$Q_{\text{Mandel}} > 0$: Super-Poissonian (bunched) statistics.

$Q_{\text{Mandel}} < 0$: Sub-Poissonian (antibunched) statistics.

Under filtered illumination: $Q_{\text{Mandel}} = 0.34(5)$ (enhanced bunching).

References

- [1] Adolphs, J.; Renger, T. Biophys. J. **2006**, *91*, 2778–2797.
- [2] Engel, G. S.; Calhoun, T. R.; Read, E. L.; Ahn, T.-K.; Mančal, T.; Cheng, Y.-C.; Blankenship, R. E.; Fleming, G. R. Nature **2007**, *446*, 782–786.
- [3] Panitchayangkoon, G.; Hayes, D.; Fransted, K. A.; Caram, J. R.; Harel, E.; Wen, J.; Blankenship, R. E.; Engel, G. S. Proc. Natl. Acad. Sci. U.S.A. **2010**, *107*, 12766–12770.

Table 7: **Computational Performance**

Task	CPU Time
Single trajectory (1000 fs)	31.2 min
100 disorder realizations	52.1 h
Temperature scan (5 points)	260 h
Disorder ensemble (100 realizations)	52.1 h
Bath parameter scan (9 points)	469 h
Total for manuscript	1 840 h

- [4] Ishizaki, A.; Fleming, G. R. Proc. Natl. Acad. Sci. U.S.A. **2009**, *106*, 17255–17260.
- [5] Collini, E.; Wong, C. Y.; Wilk, K. E.; Curmi, P. M. G.; Brumer, P.; Scholes, G. D. Nature **2010**, *463*, 644–647.
- [6] Mohseni, M.; Rebentrost, P.; Lloyd, S.; Aspuru-Guzik, A. J. Chem. Phys. **2008**, *129*, 174106.
- [7] Rebentrost, P.; Mohseni, M.; Kassal, I.; Lloyd, S.; Aspuru-Guzik, A. New J. Phys. **2009**, *11*, 033003.
- [8] Huelga, S. F.; Plenio, M. B. Contemp. Phys. **2013**, *54*, 181–207.
- [9] Christensson, N.; Kauffmann, H. F.; Pullerits, T.; Mančal, T. J. Phys. Chem. B **2012**, *116*, 7449–7454.
- [10] Suess, D.; Eisfeld, A.; Strunz, W. T. Phys. Rev. Lett. **2014**, *113*, 150403.
- [11] Citty, B.; Lynd, J. K.; Gera, T.; Varvelo, L.; Raccach, D. I. G. B. J. Chem. Phys. **2024**, *160*, 144105.
- [12] Chen, L.; Bennett, D. I. G.; Eisfeld, A. J. Chem. Phys. **2022**, *156*, 124101.
- [13] Varvelo, L.; Lynd, J. K.; Bennett, D. I. G. Chem. Sci. **2021**, *12*, 9704–9711.
- [14] Scholes, G. D.; Fleming, G. R.; Olaya-Castro, A.; van Grondelle, R. Nat. Chem. **2011**, *3*, 763–774.
- [15] Moix, J. M.; Wu, J.; Hu, P.; Cao, J. J. Chem. Phys. **2011**, *134*, 164106.
- [16] Baumgratz, T.; Cramer, M.; Plenio, M. B. Phys. Rev. Lett. **2014**, *113*, 140401.
- [17] Pezzé, L.; Smerzi, A. Phys. Rev. Lett. **2009**, *102*, 100401.
- [18] Jang, S.; Cheng, Y.-C.; Reichman, D. R.; Eaves, J. D. J. Chem. Phys. **2008**, *129*, 101104.